

Nicolò Marmi

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EDUCATION

Stony Brook University | Stony Brook, NY

August 2022-December 2026

Honors Bachelor of Science, Computer Science Major, Applied Mathematics & Statistics Major

- Cumulative **GPA: 3.91/4.0** | Dean's List Scholar | Presidential Merit Scholarship
- Computer Science Honors Program

Relevant Coursework: Data Structures and Algorithms, Honors Analysis of Algorithms, Machine Learning, Software Engineering, Finite Mathematical Structures, Probability and Statistics, Data Analysis.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, C, C++, OCaml, SQL

Frameworks & Libraries: PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, React, Next.js, Express.js, TA-Lib, Polars

Tools & Platforms: Git, Docker, MongoDB, REST APIs, Linux/Unix, AWS (EC2/S3), Vercel

WORK EXPERIENCE

Fineco Asset Management

Dublin, Ireland

(Incoming) Quantitative Developer Intern

May 2026-August 2026

- Selected for the quantitative development team to build NLP pipelines that generate trading signals.

Capco

Milan, Italy

Software Engineer Intern

April 2024-October 2024

- Built the translation layer (OpenAI API, Pydantic validation) for a pipeline enabling UniCredit to collect branch-employee feedback across ~2,000 branches for senior management.
- Redesigned the post-processing module mapping unstructured model output onto a complex hierarchical tag schema (nested topic tags several levels deep), replacing an $O(N^2)$ approach with a two-pass $O(N)$ algorithm.
- Identified and eliminated redundant system prompts (~10% of survey tokens), significantly reducing API costs.
- Refactored modules of LORE ML explainability framework, improving code clarity and maintainability. [\[1\]](#)

PROJECTS & ACTIVITIES

Quantum ML

Stony Brook, NY

Honors Research Thesis

January 2026-December 2026

- In collaboration with Brookhaven National Laboratory, designed a two-stream neural network in PyTorch to predict unknown reduced density matrices from known marginals of the same quantum system.
- Reconstructed quantum states from partial measurements, reducing tomography cost and enabling more efficient quantum simulations.

DraftIQ + Player Data API | *JavaScript, Node.js, Next.js, MongoDB Atlas, Git*

Stony Brook, NY

Creator

January 2026-May 2026

- Led a team of 4 students building DraftIQ, a full-stack fantasy baseball draft application, and a player data API providing MLB statistics, analytics, and draft recommendations.
- Engineered a scalable backend using Express and MongoDB aggregation pipelines with async request handling, optimized for low-latency queries.

Algorithmic Trading & Backtesting System | *Python, Polars, TA-Lib, Git*

Remote

Software Engineer, Algorithms Engineer

August 2024-February 2025

- Engineered trading algorithms adapting strategies from two algorithmic trading books. [\[2\]](#), [\[3\]](#)
- Designed and implemented a modular, reusable trading system with a backtesting framework, enabling replicable experiments and easy extension of new strategies.
- Validated strategies through out-of-sample backtesting and paper trading, with transaction costs modeled via bid-ask spread to ensure realistic performance estimates.

Clip Extractor | *Python, face-recognition, OpenCV, Git*

Milan, Italy

Creator

September 2024

- Built a Python tool to automate video sub-clip extraction via face recognition, integrating MoviePy into a scalable workflow reducing manual editing to nearly 0%.
- Implemented batch-processing logic to optimize memory usage and handle large video files efficiently.